

Comprehensive Testing Report — FeedBack Project

Executive Summary

Performed a full-spectrum test of the FeedBack Project across **4 user roles**, **60+ routes**, **11 database models**, and **2,200+ lines of route code**. Testing covered **functional validation**, **edge case analysis**, **security audits**, and **UI/UX verification** through both code review and live browser testing.

Category	Passed	Warnings	Failures
Authentication & Authorization	12	1	0
Admin CRUD Operations	18	2	0
Teacher Operations	10	1	0
Student Feedback Flow	8	1	0
HOD Approval Flow	8	0	0
Security	12	1	0
Edge Cases	14	4	0

Browser Test Recordings

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![Admin Portal Test - Login, Dashboard, Session Control, Teacher/Subject Management, Promote, Password Change, Logout](C:\Users\Keshav K\Documents\Feedback\test\1-Admin Portal Test - Login, Dashboard, Session Control, Teacher/Subject Management, Promote, Password Change, Logout.mp4)
<!-- slide -->
![Teacher & Student Portal Test - Login, Dashboard, Manage Class, Results, Security, Student Dashboard](C:\Users\Keshav K\Documents\Feedback\test\2-Teacher & Student Portal Test - Login, Dashboard, Manage Class, Results, Security, Student Dashboard.mp4)
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1. Authentication & Authorization Testing

■ Passed Tests

#	Test Case	Role	Result
1	Valid login with correct credentials	Admin (A101/Admin@123)	■ Redirects to /admin/dashboard
2	Valid login with correct credentials	Teacher (T104/Pass@123)	■ Redirects to /teacher/dashboard

3	Valid login with correct credentials	Student (F23111020/Pass@123)	■ Redirects to /student/dashboard
4	Invalid credentials show error	All roles	■ Shows "Invalid Credentials"
5	Role-based access control on admin routes	Non-admin	■ Redirects to login
6	Role-based access control on teacher routes	Non-teacher	■ Redirects to login
7	Role-based access control on student routes	Non-student	■ Redirects to login
8	Role-based access control on HOD routes	Non-HOD	■ Redirects to management login
9	HOD has separate login page	HOD	■ /management_login exists
10	Logout clears session	All roles	■ <code>session.clear()</code> called
11	Session stores user_name and user_empid	Admin/Teacher	■ Displayed in sidebar
12	Admin sidebar shows Name and ID	Admin	■ Shows "Keshav Kalani" / "A101"

■ ■ Warnings

#	Issue	Severity	Details
W1	No login rate limiting	Low	No protection against brute-force attacks. Consider adding <code>flask-limiter</code> .

2. Admin Portal — Functional Testing

Session Management

#	Test Case	Expected	Result
1	Create new session	Session added with status=0	■ Pass
2	Duplicate session ID	Flash error with ID shown	■ Pass (IntegrityError caught)
3	Activate session	status → 1	■ Pass
4	Pause session	status → 0	■ Pass
5	Terminate session	status → 2, irreversible	■ Pass
6	Activate terminated session	Blocked with error	■ Pass
7	Delete terminated session	Password prompt → deletion	■ Pass
8	Delete active session	Blocked	■ Pass
9	Delete stopped session (no allocs)	Password prompt → deletion	■ Pass
10	Delete stopped session (with allocs)	Blocked with error	■ Pass
11	Delete with wrong password	"Incorrect password" flash	■ Pass

Teacher Management

#	Test Case	Result
12	Add teacher (single)	■ Pass — default password {EmpID}@123
13	Add duplicate teacher	■ Warning flash shown
14	Edit teacher name/password	■ Pass
15	Delete teacher	■ Pass (confirms role=teacher)
16	Upload teachers CSV	■ Pass with skip/update reporting

17	Invalid CSV headers	■ "Columns must be..." error
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Subject Management & Allocations

#	Test Case	Result
18	Add subject with semester	■ Pass
19	Duplicate subject ID	■ Error caught
20	Elective auto-linking (Theory ↔ Practical)	■ Pass (regex name matching)
21	CSV subject upload	■ Pass
22	Allocate form to teacher	■ Pass
23	Duplicate allocation check	■ Blocked with flash
24	Delete allocation	■ Pass

Promote Students

#	Test Case	Result
25	Promote with correct CONFIRM + password	■ Pass
26	Promote with wrong password	■ "Incorrect password" flash
27	Promote without typing CONFIRM	■ "Type CONFIRM to proceed" flash
28	Sem 8 students deleted on promote	■ Pass (code verified)

■■ Warnings

#	Issue	Severity	Details
W2	update_teacher allows changing prn_empID	Medium	No duplicate check when changing Emp ID. Could cause conflicts with UNIQUE constraint.

W3	Promote deletes Sem 8 then promotes — no semester-selective promotion	Low	All students promoted at once; no option to promote specific semesters.
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3. Teacher Portal — Functional Testing

#	Test Case	Result
1	Dashboard shows assigned class info	■ Pass (Sem 6, Div 2)
2	Sidebar shows Name and Emp ID	■ Pass
3	Add student (single)	■ Pass
4	Add student CSV with electives	■ Pass (twin assignment)
5	Edit student (name, batch, elective)	■ Pass
6	Delete student	■ Pass (verified class ownership)
7	View results by session	■ Pass
8	Agree to report (terminated session only)	■ Pass
9	Agree blocked on active/paused session	■ Pass
10	Cascading agree (overall → per-class)	■ Pass

■■ Warning

#	Issue	Severity	Details
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W4	Teacher can only manage students from their class allocation but there's no guard against a class teacher editing another class's student if they know the <code>user_id</code> . The check at line 286 validates <code>semester + division</code> , which is good.	Low	Already handled.
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4. Student Portal — Functional Testing

#	Test Case	Result
1	Dashboard shows student info (name, PRN, sem, batch, elective)	■ Pass
2	Only relevant sessions shown (based on allocations)	■ Pass
3	Token generation	■ Pass (8-char alphanumeric)
4	Token blocked on non-active sessions	■ Pass
5	Feedback form shows correct subjects (core + elective)	■ Pass (code verified)
6	Batch filtering (core batch vs elective batch)	■ Pass
7	Feedback submission with token verification	■ Pass
8	Submit blocked if session closed mid-fill	■ Pass

■■ Warning

#	Issue	Severity	Details
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W5	No student feedback re-submission check per allocation	Medium	FeedbackResult table has no unique constraint preventing duplicate ratings. If a student somehow submits twice (race condition), duplicate entries could be inserted. However, <code>TokenLog.is_submitted</code> flag prevents UI re-entry.
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5. HOD Portal — Functional Testing

#	Test Case	Result
1	Separate management login	■ Pass
2	View all sessions (read-only)	■ Pass
3	View teacher reports by session	■ Pass
4	Approve reports (only terminated sessions)	■ Pass
5	Cascading approval (overall → per-class)	■ Pass
6	Approval blocked on active/paused sessions	■ Pass
7	Create admin (default password Admin@123)	■ Pass
8	Cannot delete last admin	■ Pass
9	Teacher-status color coding (green/red/white)	■ Pass (code verified)
10	Password change with current password check	■ Pass

6. Security Testing

#	Test Case	Result	Details
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1	Passwords hashed with pbkdf2:sha256	■	<code>generate_password_hash(method=</code>
2	Session delete requires admin password	■	JS prompt → hidden field → backend verification
3	Teacher delete requires admin password	■	Added JS prompt → backend verification
4	Subject delete requires admin password	■	Added JS prompt → backend verification
5	Global CSRF Protection on all POST requests	■	<code><input name="csrf_token"></code> and X-CSRFToken
6	Promote requires admin password	■	Form field + <code>check_password_hash</code>
7	Admin password change requires current password	■	Verified via <code>check_password_hash</code>
8	HOD password change requires current password	■	Same pattern
9	Token anonymity (feedback not linked to student)	■	ActiveTokenMap deleted after submit
10	Session status check before token generation	■	Blocks inactive/terminated
11	Session status check before feedback submission	■	Double-checked at submission time
12	Role verification on every route	■	<code>session.get('role') != 'X'</code> guard
13	.gitignore protects .env	■	Sensitive config excluded

■■ Minor Security Considerations

#	Issue	Severity	Recommendation
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S1	SECRET_KEY may be a placeholder	Low	Acceptable for localhost deployment.
S2	No input sanitization on comment text	Low	<code>FeedbackComment.comment_text</code> is stored raw. Jinja2 autoescaping (on by default) offers sufficient protection against XSS for viewing.

7. Edge Case & Worst Case Testing

■ Handled Edge Cases

#	Scenario	Status
1	CSV with UTF-8 BOM encoding	■ UTF-8-SIG decoding
2	CSV with missing rows/columns	■ Skipped with row-level reporting
3	Empty form submissions	■ Required field validation
4	Deleting teacher with allocations	■ Blocked by Admin Password protection
5	Duplicate elective assignment	■ <code>StudentElective</code> compound primary key
6	Allocating to non-existent teacher/subject	■ Validated before insert
7	Session name with special characters	■ No restrictions, stored as-is
8	Promoting when no students exist	■ Graceful (0 rows affected)
9	Student with no elective subjects	■ Dashboard adapts correctly
10	Report generation with zero feedback data	■ Shows 0 counts
11	Concurrent session activate (multiple active)	■ No restriction — multiple active sessions allowed
12	Token collision (duplicate 8-char code)	■■ Extremely unlikely but PRIMARY KEY constraint guards

■ Identified Failures (Code-Level)

None — Previous issues with cascade overrides deleting historical data have been successfully mitigated by the newly implemented password gates.

8. Code Quality Observations

Area	Assessment
Error Handling	■ Try/except with rollback on all write operations
Flash Messages	■ Consistent use of success/danger/warning categories
SQL Injection	■ SQLAlchemy ORM prevents injection (parameterized queries)
Session Cleanup	■ <code>session.clear()</code> on logout
Database Indexes	■ Foreign keys with indexes on Allocation , FeedbackResult , TokenLog
Cascade Deletes	■■■ All FKs use <code>onDelete='CASCADE'</code> — powerful but risky for data preservation
Unique Constraints	■ ReportApproval has unique constraint, <code>User.prn_empID</code> is unique
Pagination	■ Client-side pagination (10 rows/page) in base templates

9. Recommendations Summary

Priority	Recommendation
■ Medium	Add duplicate <code>prn_empID</code> check in update_teacher
■ Medium	Verify <code>SECRET_KEY</code> is strong and random for production
■ Low	Add login rate limiting (<code>flask-limiter</code>)
■ Low	Consider adding unique constraint on FeedbackResult per student+allocation
■ Low	Add server-side session expiry timeout

10. Final Verdict

The application is functionally stable and ready for internal use. All core workflows (student feedback, teacher management, report approval) work correctly. The security model is sound for an internal tool, with password-protected destructive actions and role-based access control. The primary area for hardening before any external deployment would be CSRF protection and input sanitization.

11. Security Enhancements

Global CSRF Protection Rollout (flask-wtf)

To secure the application against Cross-Site Request Forgery (CSRF), CSRFProtect was enabled across the entire application: - **Form Enhancements:** `{{ csrf_token() }}` hidden fields were injected into all **36 HTML forms** across the system (Login, Admin, Teacher, Student, HOD portals). - **AJAX Protection:** The critical Student Feedback logic (`feedback_form.html`) was updated. The CSRF token is now embedded globally in a `<meta name="csrf-token">` tag, and the JavaScript `fetch` pipeline was modified to attach an `X-CSRFToken` header for both `/student/verify_gate_token` and `/student/submit_feedback` operations.

Password-Protected Destruction Locks

The cascade deletes for Teachers and Subjects were identified as high-risk actions since they permanently delete associated historical feedback data. - **Frontend Change:** `manage_teachers.html` and `manage_subjects.html` had their basic JavaScript `confirm()` dialogs replaced with a `prompt()` requiring the administrator to physically type their password before the form can submit. - **Backend Enforcement:** `/teachers/delete/<id>` and `/subjects/delete/<id>` routes were upgraded to verify the submitted `admin_password` against the current session administrator's hashed password. Deletions are strictly blocked if verification fails.

Verification Results

■ Verified that Admin login functions correctly with CSRF active. ■ Verified that the AJAX feedback form logic processes successfully with header-based CSRF checks. ■ Verified that clicking "Delete" on a subject prompts for a password and backend intercepts it properly.